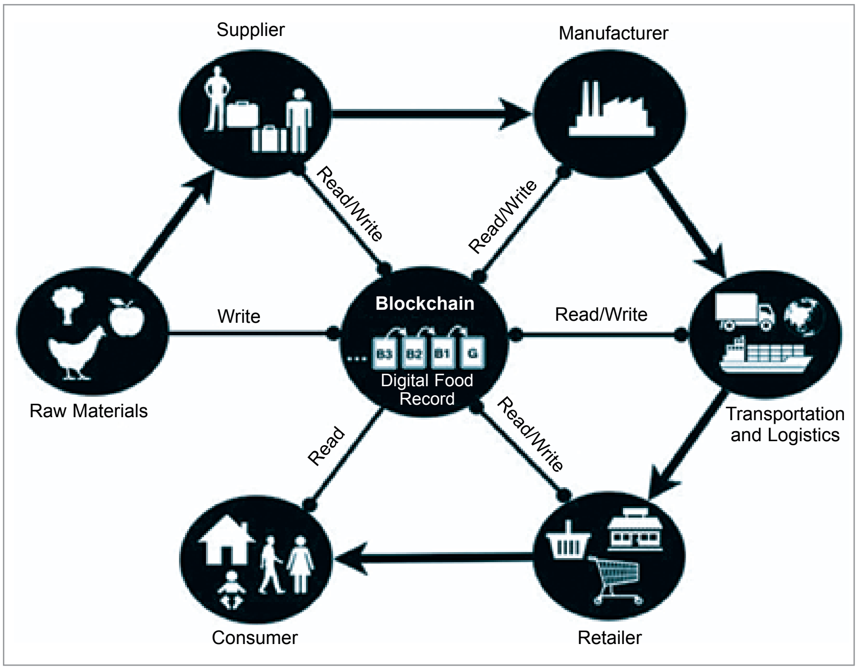

Fig. 2: Flow diagram

pass off cheaper products for more expensive varieties. Occasionally, it is a different variety altogether. Horse meat scandal and fish meat scandal in European Union during 2016 are among the many examples.

It took the United States Food and Drug Administration more than two months to identify the source of Salmonella-tainted Maradol papayas, which have thus far sickened more than 200 people in 23 states, resulting in 65 hospitalisations and at least one death. Had blockchain been used to create a digital ledger of the distribution chain, the farm in southern Mexico could have been identified within a matter of seconds.

What's keeping food fraud alive?

Here are some reasons behind the problem:

The food supply chain runs on outdated practices. Food fraud has increased by 60 per cent over the last couple of years due to lack of transparency, accountability, and effective regulations. Though, recently, regulators are demanding state-of-the-art practices and technology to help bring organisations up to the required standard and ultimately create a more transparent food system.

Complex supply chains create blind spots. Many companies simply lack the awareness of where and how they are susceptible to food fraud. However, with up to ten percent of the food system affected by food fraud, weak links can occur across raw materials, ingredients, products, and packaging.

Vulnerable regulatory systems enable tampering. Current practices of storing compliance data either on paper or in centralised databases are susceptible to inaccuracies, hacking, and intentional errors motivated by corruption.

Blockchain

In principle, a blockchain should be considered as a distributed append-only timestamped data structure. Blockchains allow us to have a distributed peer-to-peer network where non-trusting members can verifiably interact with each other without the need for a trusted authority (Christidis and Devetsikiotis, 2016).

Blockchain for food and beverages

A shared digital food supply chain powered by blockchain enables full transparency by digitising transaction records and storing them in a decentralised and immutable manner, eliminating opportunity for fraud across the food chain. "Blockchain technology enables a new era of end-to-end transparency in the global food system—equivalent to shining a light on food ecosystem participants that will further promote responsible actions and behaviours," says Frank Yiannas, Walmart's vice president for food safety. "It also allows all participants to share information rapidly and with confidence across a strong trusted network. This is critical to ensuring that the global food system remains safe for all," he adds.

Some advantages of adopting blockchain technology in food supply chain are:

End-to-end traceability. Increased surveillance shines a light on each link in the food chain, enabling real-time traceability of food fraud culprits—and creating accountability.

Collaboration. Secure data-sharing between food chain actors eliminates the possibility for participants to move fraudulent foods unknowingly.

Transparency. Improved transparency allows fewer opportunities for fraudsters to penetrate the supply chain, and permanent records enable better management of material safety and quality standards.

Blockchain and IoT, each of which bring their respective strengths, continue to be adopted by the food and beverage industry. Over the past few months, a variety of players, including giants like Nestlé and Carrefour, have reported their blockchain-powered initiatives within the field.

The most adopted blockchain tracking solution within the field is IBM's Food Trust, which is based on the Hyper Ledger Fabric blockchain protocol. The platform went live in October 2018, since when millions of individual food products have reportedly been tracked by retailers and suppliers using the Food Trust blockchain. Most recently, it was reported that salmon farming company Cermaq and smoked salmon producer Labeyrie were using IBM's cloud blockchain technology to trace their product supply chains.

The blockchain modules are designed to help participants in the